INCH-POUND

MIL-A-82950(OS)
9 June 1995

MILITARY SPECIFICATION

# ACTUATOR, BELLOWS ATU-117/P

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the ATU-117/P Bellows Actuator. For the purposes of this specification the term "bellows actuator" or "actuator" shall refer to the ATU-117/P Bellows Actuator (see 6.1).

## 2. APPLICABLE DOCUMENTS

**2.1 Government documents.**

**2.1.1 Specifications and standards.** The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-A-430 | Aluminum Alloy Rod and Wire: For Rivets and Cold Heading |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner |

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-D-23615 | Design and Evaluation of Cartridge Actuated Devices |
| MIL-I-23659 | General Design Specification for Electric Initiators |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

**DISTRIBUTION STATEMENT A.** Approved for public release; distribution is unlimited.

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-453 | Inspection, Radiographic |
| MIL-STD-810 | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-1168 | Lot Numbering of Ammunition |

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, Attn: NPODS, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

**2.1.2 Other Government documents, drawings and publications.** The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Air Systems Command (Code 30003)

| | |
|---|---|
| 837AS266 | Actuator, Bellows ATU-117/P |

Pacific Scientific

| | |
|---|---|
| 2-951024 | Test Fixture |

(Application for copies of drawings should be addressed to Commander, Naval Surface Warfare Center, Data Control Branch (Code 8410), Indian Head Division, Indian Head, MD 20640-5035.)

CODE OF FEDERAL REGULATIONS (CFR)

| | |
|---|---|
| 49 CFR 100-199 | Transportation |

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

**2.2 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).



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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 5118 Fiberboard Shipping Boxes, Standard Practice for Fabrication of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

**2.2 Order of precedence.** In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified in the contract or purchase order, a sample of bellows actuators shall be subjected to first article inspection (see 4.4 and 6.3)

**3.2 Flight critical part.** The bellows actuator is a flight critical item and the technical data package shall be identified and procured as such.

**3.3 Design and construction.** The following applies to the qualification status and configuration control of the bellows actuator over the entire life of the item.

**3.3.1 Identical configuration.** The bellows actuator design and construction shall be identical to that subjected to qualification testing and approved for service use by the Navy. The bellows actuator installed life shall be at least 36 months in the operational environment of −20 to 120°F. The shelf life shall be at least 60 months, as calculated from the lot number.

**3.3.2 Configuration control.** Only bellows actuators procured from the vendor(s) listed on drawing 837AS266 are approved for the application specified. A substitute bellows actuator shall not be used without prior approval and testing by the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5035 and the Naval Air Systems Command Headquarters.

**3.3.3 HERO.** The bellows actuator shall be designed to pass the Hazards of Electromagnetic Radiation to Ordnance test.

**3.4 Bellows actuator production.** Bellows actuators shall be manufactured in accordance with Drawing 837AS266 and this specification. Each production bellows actuator shall meet the requirements of 3.4.1 through 3.4.6 as outlined in table II. Failure of any bellows actuator to meet these requirements shall result in rejection of that part. The explosive for each production lot shall be from the same explosive lot.

**3.4.1 Visual inspection.** Bellows actuators shall be free of the following defects: illegible, missing or inaccurate identification markings, damage, burrs, dents, sharp edges, or other defects which may prevent installation of the bellows actuator into the wedge/track assembly (see 4.3.1).



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**3.4.2 Radiographic inspection.** When radiographically examined in accordance with 4.3.2, the bellows actuator shall show proper assembly, presence of parts, sealing, and proper electrical connections. The radiographic inspection shall show no foreign materials present.

**3.4.3 Leakage.** The leakage rate of each bellows actuator shall not exceed $1.0 \times 10^{-5}$ cc/sec of air when tested in accordance with 4.3.3.

**3.4.4 Bridgewire resistance.** Bellows actuators shall have a bridgewire resistance of $1.0 \pm 0.1$ ohms when tested without the cable assembly attached and shall have a resistance of $1.1 \pm 0.2$ ohms with the cable assembly attached, in accordance with 4.3.4

**3.4.5 Electrostatic discharge.** Bellows actuators shall not initiate or incur any internal/external damage or degradation as a result of the electrostatic discharge test of 4.3.5.

**3.4.6 Insulation resistance.** Each bellows actuator shall have a resistance greater than 1 megohm between each conductor and every other isolated circuit in the assembly when tested in accordance with 4.3.6.

**3.4.7 One amp/one watt no-fire.** Bellows actuators shall not initiate or incur any internal/external damage or degradation as a result of the one amp/one watt no-fire test of 4.3 7.

**3.5 Bellows actuator performance.** Actuators shall meet all requirements of 3.5.1 through 3.5.7.

**3.5.1 Shock.** Bellows actuators shall not initiate or incur any internal/external damage or degradation when a saw tooth shock pulse is applied in accordance with 4.3.8.

**3.5.2 Temperature/humidity/altitude (THA) cycling.** Bellows actuators shall not initiate or incur any internal/external damage or degradation when a THA cycle is applied in accordance with 4.3.9.

**3.5.3 Low temperature.** Bellows actuators shall not initiate or incur any internal/external damage or degradation when tested in accordance with 4.3.10.

**3.5.4 Vibration.** Bellows actuators shall not initiate or incur any internal/external damage or degradation when vibration cycles are applied in accordance with 4.3.11.

**3.5.5 Salt fog.** Bellows actuators shall not initiate or incur any internal/external damage or degradation as a result of the salt fog test of 4.3.12.

**3.5.6 Six-foot drop.** Bellows actuators shall not initiate or incur any major internal/external damage or degradation as a result of the six-foot drop test of 4.3.13.

**3.5.7 Forty-foot drop.** Bellows actuators shall not initiate or incur any damage or degradation which makes them unsafe for handling or disposal as a result of the forty-foot drop test of 4.3.14.

**3.5.8 Functional performance requirements.** Bellows actuators shall meet the following functional performance requirements of 4.3.15.



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**3.5.8.1 Release.** The bellows actuator shall overcome a 40-pound load and expand a minimum of 1.00 inch in the test fixture to simulate complete release of the goggles.

**3.5.8.2 Ruptures.** The bellows actuator must demonstrate the ability to function without rupturing.

**3.5.8.3 Function time.** The upper limit (UL) for function time (application of current to 1 inch travel) over the temperature range −20 to 120°F shall be 15.0 milliseconds. The lot shall be acceptable with respect to function time if each actuator in the test sample meets the function time requirement. The acceptable number of defects is zero and the rejection number is one or more.

**3.5.8.4 System temperatures.** The bellows actuator shall be designed for a storage/environmental temperature range of −65° to 165°F. The operating/functional temperature range for the Cat's Eyes Emergency Detachment System (CEEDS) is −20° to 120°F.

**3.5.8.5 Design.** The bellows actuator shall be a self-contained device such that no fragmentation or release of flame, smoke or gas shall occur.

**3.6 System safety.** The design, construction, and operation of the bellows actuator shall present no hazard to personnel and equipment during normal transportation, storage, handling, or use. Safety provisions shall reflect full consideration of potential sources of personnel injury and equipment damage.

**3.7 Workmanship.** Bellows actuators shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes, sealing, and assembly operations.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.1.1 Responsibility for compliance.** All items shall meet the requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.



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**4.2 Classification of inspections.** The inspection requirements specified herein are as follows:

a. First article inspection (see 4.2.1)
b. Production inspection (see 4.2.2)
c. Lot acceptance inspection (see 4.2.3)

**4.2.1 First article inspection.** When more than 36 months have elapsed since production of the latest accepted lot, a sample of bellows actuators shall be subject to first article testing. Subject the items to the tests described in table I. The first article sample shall be produced using the same procedures and methods proposed for the production lot. Unless otherwise specified, 24 bellows actuators representative of the planned production lot shall be sequentially tested, temperature conditioned, and functionally tested. The first article sample shall not be applied as part of the quantity specified for delivery by the contract. Any production lot items manufactured prior to notification by IHDIVNSWC of first article sample acceptability shall be done at the contractor's risk.

**4.2.2 Production inspection.** All production bellows actuators manufactured under contract shall be inspected and screened for defects. Bellows actuators failing to meet the requirements listed in table II shall be rejected and removed from the lot. In addition, all packaging shall meet the requirements of section 5. All discrepancies shall be corrected before acceptance.

TABLE I. First article test schedule.

| Test | Para. | A | B | C | D | E | F |
|---|---|---|---|---|---|---|---|
| Visual Inspection | 4.3.1 | 9 | 3 | 3 | 3 | 3 | 3 |
| Radiographic Examination | 4.3.2 | 9 | 3 | 3 | 3 | 3 | 3 |
| Leakage | 4.3.3 | 9 | 3 | 3 | 3 | 3 | 3 |
| Bridgewire Resistance | 4.3.4 | 9 | 3 | 3 | 3 | 3 | 3 |
| Electrostatic Discharge | 4.3.5 | 9 | 3 | 3 | 3 | 3 | 3 |
| Insulation Resistance | 4.3.6 | 9 | 3 | 3 | 3 | 3 | 3 |
| 1 Amp 1 Watt | 4.3.7 | 9 | 3 | 3 | 3 | 3 | 3 |
| Shock | 4.3.8 | 9 | | | | | |
| THA | 4.3.9 | 9 | | | | | |
| Low Temperature | 4.3.10 | 9 | | | | | |
| Vibration | 4.3.11 | 9 | | | | | |
| Salt Fog | 4.3.12 | 9 | | | | | |
| Six-Foot Drop | 4.3.13 | | 3 | | | | |
| Forty-Foot Drop | 4.3.14 | | | 3 | | | |
| Visual Inspection | 4.3.1 | 9 | | | | | |
| Radiographic Examination | 4.3.2 | 9 | | | | | |
| Functional Performance @ −20°F | 4.3.15 | 3 | | | 3 | | |
| Functional Performance @ 70°F | 4.3.15 | 3 | 3 | | | 3 | |
| Functional Performance @ 120°F | 4.3.15 | 3 | | | | | 3 |



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TABLE II. Production inspections.

| Test Sequence | Test Paragraph | Requirement Paragraph |
|---|---|---|
| 1. Visual Inspection | 4.3.1 | 3.4.1 |
| 2. Radiographic Inspection | 4.3.2 | 3.4.2 |
| 3. Leakage | 4.3.3 | 3.4.3 |
| 4. Bridgewire Resistance | 4.3.4 | 3.4.4 |
| 5. Electrostatic Discharge | 4.3.5 | 3.4.5 |
| 6. Insulation Resistance | 4.3.6 | 3.4.6 |
| 7. Packaging | 4.2.2 | Section 5 |

**4.2.3 Lot acceptance inspection.** Lot acceptance inspection shall consist of the inspections and tests specified in table III. Failure of any sample bellows actuator to comply with the requirements listed in table III shall be cause for rejection of the lot represented.

TABLE III. Lot acceptance inspections and tests.

| Inspection/Test | Test Paragraph | Requirement Paragraph | Quantity |
|---|---|---|---|
| Visual Inspection | 4.3.1 | 3.4.1 | Test sample |
| Radiographic Inspection | 4.3.2 | 3.4.2 | Test sample |
| 1 Amp 1 Watt No-Fire | 4.3.7 | 3.4.7 | Test sample |
| Functional Test:<br>−20°F<br>70°F<br>120°F | 4.3.15 | 3.5.7 | <br>1/3 of test sample<br>1/3 of test sample<br>1/3 of test sample |

**4.2.3.1 Lot size.** Unless otherwise specified, the bellows actuator production lot size shall be from 10 minimum to 650 maximum. Bellows actuators required for all test purposes shall be randomly selected from the production lot and will not be applied as a part of the quantity specified for delivery.



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TABLE IV. Lot acceptance test sample size for given lot size.

| Lot Size | LAT Sample Size | Test | Retain |
|---|---|---|---|
| 10-50 | 10 | 9 | 1 |
| 50-100 | 20 | 18 | 2 |
| 101-300 | 30 | 27 | 3 |
| 301-650 | 50 | 45 | 5 |

**4.3 Test requirements.**

**4.3.1 Visual inspections.** All bellows actuators manufactured under contract shall be visually examined. Those with illegible or inadequate descriptive markings or serial numbers, dents, deep scratches, or any other defects which may prevent proper installation or functioning shall be rejected. Visual defects found during non-destructive inspections and destructive tests will result in 100% screening of the production lot at the contractor's expense.

**4.3.2 Radiographic examination.** Subject the specified number of bellows actuators manufactured under contract to radiographic examination. Perform radiographic examination in accordance with MIL-STD-453. Any observable imperfections in assembly shall be cause for rejection. The item shall be positioned for the most revealing exposure. All items shall be identified with serial numbers prior to examination. The items for examination shall be arranged on trays or boards in consecutive numerical order by serial number and each radiographic plate shall carry a permanent identification of the items displayed thereon. The radiographic plate identification shall include the nomenclature, ATU-117/P Bellows Actuator, the lot number as stamped on the items, and the span of serial numbers displayed. Radiographic plates of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance or first article test program.

**4.3.3 Leakage.** Subject the specified number of bellows actuators to leakage testing. The bellows actuators shall not exhibit a leak rate greater than $1.0 \times 10^{-5}$ cubic centimeters per second of air at standard temperature and pressure. Any test item which exhibits a leak rate in excess of the calculated value at a pressure differential of $1.0 \pm 0.1$ atmosphere shall be considered defective.

**4.3.4 Bridgewire resistance.** The bridgewire resistance shall be measured using a direct current of 5 to 10 milliamperes. The bridgewire resistance shall be 0.9 to 1.1 ohms for all actuators when measured without cable assemblies attached. When the cable assemblies are attached, the bridgewire resistance shall measure 0.9 to 1.3 ohms.

**4.3.5 Electrostatic discharge.** Subject the specified number of bellows actuators to the electrostatic discharge test in accordance with MIL-I-23659. The latest revision applies. Perform the test in two configurations: between the two leads of the actuator, and between the two leads shorted together and the case of the actuator. The actuators shall not fire or dud when subjected to the simulated 25,000-volt human electrostatic discharge.



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**4.3.6 Insulation resistance.** The insulation resistance shall be measured between the leads of the bellows actuator shorted together and the case. The insulation resistance measurement shall be made and the actual current recorded immediately after application of uninterrupted test voltage. The actuators shall have a resistance greater than 1 megohm.

**4.3.7 One amp/one watt no-fire.** A direct current of not less than one ampere supplying a minimum of 1 watt shall be applied to the bridge circuit for at least 5 minutes. The actuator shall not fire when subjected to this test.

**4.3.8 Shock.** Subject the specified number of bellows actuators to shock testing. Perform testing in accordance with MIL-STD-810E. The shock pulse wave form shall be terminal peak sawtooth. The peak amplitude shall be 20g with a duration of 11 milliseconds as shown in figure 1. Three shock pulses shall be applied in both the positive and negative direction of the X-axis, Y-axis, and Z-axis. There shall be a total of eighteen shocks. The actuators shall not fire when subjected to shock testing. Each actuator shall meet post-test inspections and functional performance requirements.

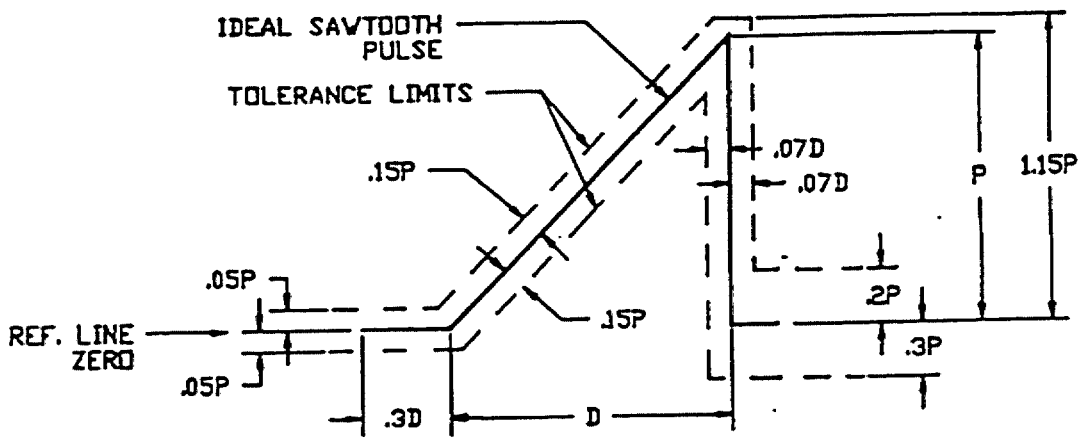



FIGURE 1. Shock pulse wave form.

**4.3.9 Temperature, humidity and altitude.** Subject the specified number of bellows actuators to THA testing. Perform testing in accordance with the requirements of MIL-I-23659. The latest revision of MIL-I-23659 applies. The actuators shall withstand the THA environment. Each unit shall meet post-test inspections and functional performance requirements.



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**4.3.10 Low temperature.** Subject the specified number of bellows actuators to low temperature testing. Place the units in a temperature conditioning chamber set to $-65 \pm 5°F$ and allow them to stabilize. Maintain the temperature and allow the units to soak for 48 hours. Then remove the units and allow them to return to ambient conditions. The actuators shall withstand the low temperature environment. Each unit shall meet post-test inspections and functional performance requirements.

**4.3.11 Vibration.** Subject the specified number of bellows actuators to vibration testing. Perform testing in accordance with MIL-STD-810E, method 514.4. The wave form is shown in figure 2. Perform testing at each axis for 1 hour at each temperature ($-65$, 70, or 165°F). The total testing time per group shall be 9 hours. Each actuator shall also meet post-test inspections and functional performance requirements.

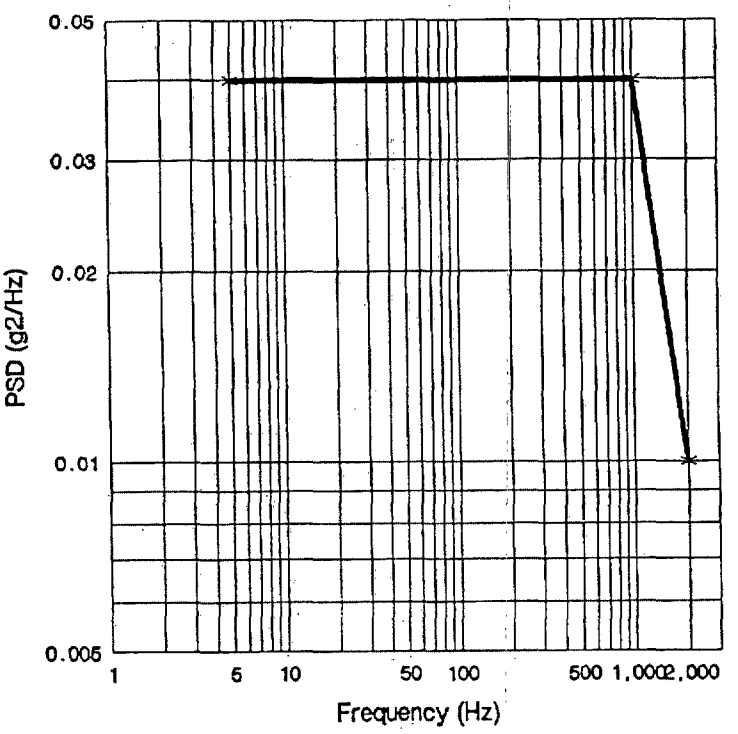



FIGURE 2. CEEDS random vibration.

**4.3.12 Salt fog.** Subject the bellows actuators to salt fog testing in accordance with method 509 of MIL-STD-810E. The salt concentration shall be a $5 \pm 1$ percent solution. The test cycle shall be as follows: salt fog condition units for 24 hours; dry at ambient conditions for 24 hours; salt fog condition units for 24 hours; dry at ambient conditions for 24 hours. There shall be a total of 48 hours of salt fog conditioning. Each unit shall meet post-test inspections and functional performance requirements.



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**4.3.13 Six-foot drop.** Subject the specified number of bellows actuators to 6-foot drop testing. Drop each item onto a 2-inch-thick steel plate embedded in concrete from a minimum height of 6 feet. Guide the items so that they impact the steel plate in the following three orientations: (1) bellows tip-end down; (2) horizontal; (3) bellows lead-end down. Use a new actuator for each drop/orientation. The actuators shall not fire when dropped from a height of 6 feet, and shall meet post-test inspections and functional performance requirements.

**4.3.14 Forty-foot drop.** Subject the specified number of bellows actuators to 40-foot drop testing. Drop each item onto a 2-inch-thick steel plate embedded in concrete from a minimum height of 40 feet. Guide the items so that they impact the steel plate in the following three orientations: (1) bellows tip-end down; (2) horizontal; (3) bellows lead-end down. Use a new actuator for each drop/orientation. The actuators shall not fire when dropped from a height of 40 feet. Each actuator shall also be safe for handling and disposal.

**4.3.15 Functional performance.** The functional test shall simulate the bellows actuator pushing a wedge under the lever bar of the Cat's Eyes Night Vision Goggles and releasing the goggles from the aviator's helmet. The test fixture shall be designed such that the bellows actuator, upon first movement, overcomes a 40-pound load, then expands a minimum of 1.00 inch. The units shall be functioned in accordance with table I or III, whichever is appropriate.

**4.3.15.1 Test fixture.** The test fixture shall be Pacific Scientific Drawing 2-951024 or equivalent. An equivalent fixture shall be an aluminum cylinder with the following features: a threaded plug to hold the bellows actuator in place, a piston placed up against the tip of the bellows actuator with a shear pin through the piston and the fixture housing, a breaklead or equivalent to measure first indication of piston movement, and a second breaklead or equivalent to measure 1.00 inch of travel. The government shall have final approval of any equivalent fixture before testing is initiated. Any testing prior to the government's approval shall be at the contractor's risk.

**4.3.15.2 Shear pin.** The shear pin shall be aluminum wire 2017-H13 per QQ-A-430, diameter 0.034 to 0.036 or equivalent to achieve a double shear force of 40 $\pm$ 10 pounds. Length shall be as required to fit the test fixture.

**4.3.15.3 Temperature conditioning.** The conditioning time shall be 1.0 hour minimum. The units may be out of the conditioning environment for a maximum of 5 minutes before functional test. If the unit is not functioned within 5 minutes, it shall be returned to the conditioning chamber for another hour of temperature conditioning.

**4.3.16 Test failure.** If a test failure is attributed to a specific cause, excluding the bellows actuator, the original test results shall be discarded and that test reconducted. Otherwise there shall be no retests.

## 5. PACKAGING

**5.1 Preservation and packaging.** The definitions of packaging levels are contained in 6.4.

**5.1.1 Level A.** Level A packaging shall be used for packaging of all production lots for service use (see 6.4.1).



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**5.1.1.1 Inner container.** One bellows actuator shall be packaged in accordance with ASTM D 5118. Place each unit into its own carton/box with compatible dunnage. A tight pack is mandatory.

**5.1.1.2 Outer container.** Bellows actuators packaged as described in 5.1.1.1 shall be packed in a nailed wood box in accordance with PPP-B-601 or PPP-B-621. Unless otherwise specified in the contract or purchase order, the number of inner containers packed in the outer container shall be at the option of the supplier. A tight pack is mandatory to prevent free movement within the outer container. Only bellows actuators having the same lot number shall be packed in a single outer container.

**5.1.2 Level C.** Bellows actuators intended for expenditure in first article and lot acceptance testing shall be packaged in accordance with the contractor's standard packaging procedures (see 6.4.3).

**5.2 Packing.** For definitions of packing levels see 6.4.

**5.2.1 Level A.** Level A packing shall be used for packing of all production lots for service use (see 6.4.2).

**5.2.2 Level C.** Bellows actuators intended for expenditure in first article sample test and for production lot acceptance tests, and packaged in accordance with 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with PPP-B-636 (see 6.4.4).

**5.3 Markings.**

**5.3.1 Normal markings.** Unless otherwise specified in the contract or purchase order the marking information of the inner and outer containers shall be as specified below:

INNER CONTAINER MARKING

a. NSN 1377-01-XXX-XXXX and DODIC XXXX
b. ATU-117/P Bellows Actuator
c. DRAWING 837AS266
d. SERIAL NUMBER
e. QUANTITY
f. LOT NUMBER IN ACCORDANCE WITH MIL-STD-1168
g. UN CLASS AND DIVISION AND COMPATIBILITY GROUP

OUTER CONTAINER MARKING

a. NSN 1377-01-XXX-XXXX and DODIC XXXX
b. ATU-117/P Bellows Actuator
c. DRAWING 837AS266
d. QUANTITY
e. GROSS WEIGHT AND CUBE
f. LOT NUMBER IN ACCORDANCE WITH MIL-STD-1168



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g. CONTRACT/PURCHASE ORDER NUMBER
h. UN CLASS AND DIVISION AND COMPATIBILITY GROUP

**5.3.2 Bar coding.** Unless otherwise specified in the contract or purchase order, the outer container shall be bar coded in accordance with MIL-STD-129-1. Additionally, the above specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129-1.

**5.4 Inspection.** The inspection of packaging requirements shall be in accordance with 4.2.2.

**6. NOTES.**

**6.1 Intended use.** The ATU-117/P Bellows Actuator is used to activate the Cat's Eyes Emergency Detachment System. At the start of an emergency ejection sequence, an electrical signal is delivered to the bellows actuator causing it to expand and push a wedge to release the night vision goggles from an aviator's helmet.

**6.2 Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether first article or qualification testing is required and, if so, specify the test activity.

d. Production lot size and lot acceptance test activity.

e. Directions for shipping radiographic plates.

f. Marking if other than as specified.

**6.3 First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products, who wish to rely on such previous production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.



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**6.4 Definitions.** For the purpose of this specification the following definitions apply.

**6.4.1 Level A packaging.** The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical handling, shipment, indeterminate storage and world-wide redistribution.

**6.4.2 Level A packing.** The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide distribution.

**6.4.3 Level C packaging.** The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level and MIL-STD-129.

**6.4.4 Level C packing.** The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first activity for immediate use. This level will conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level and MIL-STD-129.

**6.4.5 Flight critical part.** A part, the single failure of which, during any operating condition, could cause loss of aircraft or one of its major components, loss of control, unintentional release of, or inability to release, or which may cause significant personnel injury, including during flight, escape, survival, or rescue.

**6.4.6 Function time.** The time interval between initial application of current (to the bridgewire) and release of the goggles or shear pin simulation of goggles.

**6.4.7 Production lot.** A bellows actuator production lot is defined as a group of bellows actuators manufactured consecutively and continuously, and all contain propellant from the same lot.

**6.5 Subject term (keyword) listing.**

Ejection
Emergency
HERO

Preparing activity:
Navy - OS
(Project 1377-N017)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-A-82950(OS) | 950609 |

3. DOCUMENT TITLE
ACTUATOR, BELLOWS ATU-117/P

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| COMMANDER, INDIAN HEAD DIVISION<br>NAVAL SURFACE WARFARE CENTER (CODE 8420) | (1) Commercial<br>301-743-4358/4510 | (2) AUTOVON<br>354-4358/4510 |
| c. ADDRESS (Include Zip Code)<br>101 STRAUSS AVENUE<br>INDIAN HEAD, MD 20640-5035 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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